Review notes · for you

Shortest and best: unit check

Optimization · Level 1 · Unit check

Time	Materials
20 min	Paper and a pen. No device needed.

Learning goals

- Pack the truck
- The shortest way
- Puzzles with rules
- Fair turns without clashes

Activity: Puzzle stations

1. Truck (5 min): With the six loads from lesson 1, find two loads that fit in 1,000 kg and add up their points. Which is best?
2. Map (5 min): Copy the village map and use the closest-first trick to find the shortest way from home to school.
3. Sudoku (5 min): Swap a 4 × 4 sudoku with a partner and solve it by crossing out. Count your guesses.
4. Turns (5 min): Make a fair rota for a class job so everyone has one turn before anyone has two.

Do the stations alone or with a partner, on paper. No device needed.

Discussion

- Which idea from this unit will you use first?
- When did a quick way or a quick look fool you?
- How would you explain fair turns to a younger child?

Check yourself

- My favourite idea from this unit + where I'll use it.

Try it this week
Pick your favourite idea from this unit (the best load, the closest-first trick, crossing out, or fair turns) and use it this week at school or at home. Then explain it to someone younger.

Quiz sheet

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Name _____ Class _____ Date _____

1 **The truck can carry 1,000 kg. Which load is the best?**

◯ Desks and books: 1,000 kg, 65 points ◯ Solar panels, water filter and chalk: 1,000 kg, 60 points
◯ Solar panels and desks: 1,100 kg, 75 points

2 **The best load is the one with the most points, even if it is too heavy.**

◯ True ◯ False

3 **Which of these are rules: things you must keep? Choose all that apply.**

☐ No number twice in a sudoku row ☐ Two joined tests can't share a time ☐ The shortest way
☐ No more than 1,000 kg on the truck ☐ The most points

4 **Put the closest-first trick in order.**

☐ Give its neighbours smaller numbers if you can
☐ Stop when the school is settled
☐ Give home the number 0
☐ Settle the waiting village with the smallest number

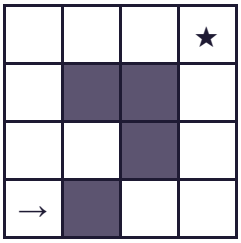
5 **The bridge between Kesi and Mina is flooded. What is the shortest way to school now?**

◯ Through Rubi and Sana: 14 km ◯ Through Bora: 15 km ◯ Through Kesi and Mina: 13 km

6 **Match each puzzle to its goal.**

Pack the truck	• • • •	The fewest kilometres
The way to school	• • • •	The fewest times
Test week	• • • •	The most points

7 **Take the robot to the star by the shortest way. Use 5 cards or fewer.**



Cut out cards (↑ forward, ↶ turn left, ↷ turn right) and lay out a program that takes the robot to the ★. Write it here:

_____

_____

8 **In a sudoku, if a square has only one candidate left, you can write it in for sure.**

◯ True ◯ False

9 **Complete the sentence.**

A _____ between two tests means they can't be at the same _____.

Word bank: time · dot · line · colour

10 **Omar's three tests are all joined to each other. What is the fewest number of times the test week can use?**

◯ 8 ◯ 3 ◯ 2

Check your answers

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1 **Desks and books: 1,000 kg, 65 points**

Solar panels and desks have the most points, but 1,100 kg breaks the rule, so they don't count.

2 **False**

The rule comes first. A load that is too heavy can't go on the truck, so it can never be the best.

3 **No more than 1,000 kg on the truck; No number twice in a sudoku row; Two joined tests can't share a time**

A rule must always be kept: the weight limit, no repeats in a row, no clashes.

4 **Top to bottom: 3, 4, 1, 2**

Start from home, always settle the smallest number, update the neighbours, and stop when the school is settled: its number is the shortest way.

5 **Through Rubi and Sana: 14 km**

The 13 km way uses the flooded bridge, so it's closed.

6 **Pack the truck — The most points**
The way to school — The fewest kilometres
Test week — The fewest times

Each puzzle has its own goal: the most points, the fewest kilometres, the fewest times.

7 **Any program that takes the robot to the ★ with 5 cards or fewer.**

The wall blocks the robot's way forward, so turn left, go up three squares, turn right and go along three: six moves.

8 **True**

Every other number has been crossed out by the rules, so the last candidate is the only number that can go there.

9 **A line between two tests means they can't be at the same time.**

A line joins two tests that one pupil takes, so they clash and need different times (colours).

10 **3**

Three tests that are all joined need three different times, so fewer than 3 is impossible, and 3 is enough for the whole week.